

CAMBRIDGE
INTERNATIONAL EXAMINATIONS

NOVEMBER 2001

ADVANCED SUBSIDIARY LEVEL

MARK SCHEME

MAXIMUM MARK : 50

SYLLABUS/COMPONENT : 8709/4

MATHEMATICS



UNIVERSITY of CAMBRIDGE
Local Examinations Syndicate

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	AS Level Examinations – November 2001	8709	4

1	WD = $30 \times 80 \cos 10^\circ$ Answer: 2360J	M1 A1 A1	For using $WD = Fd \cos \alpha$ 3
2	(i) $\frac{9500-1500}{440-120}$ Answer: 25ms^{-1} (ii) Any two of the following three features: <i>Graph starts at the origin and terminates on the t-axis</i> <i>The acceleration stage is less steep than the deceleration stage</i> <i>25ms^{-1} (f.t. for ans (i)) is correctly shown</i> All three of the above features	M1 A1 M1 A1	For attempting to find the gradient of the relevant section 2 For drawing 3 connected straight line segments with, in order, +ve, zero and -ve slopes 3
3	$R = mg \cos 30^\circ$ $F = 0.4 mg \cos 30^\circ$ Component of the weight down the plane = $mg \sin 30^\circ$ $0.4 mg \cos 30^\circ + P = mg \sin 30^\circ$ Answer: $P = 0.768$	B1 M1 M1 B1ft A1ft A1	For using $F = \mu R$ For resolving forces along the plane f.t. for cos instead of sin, following earlier cos/sin mix Depends on both M marks; f.t. for wrong F or wrong weight component 6
4	(i) $s = 2t^2 - 0.01t^4$ $2t^2 - 0.01t^4 = 100$ Answer: $t = 10$ (ii) $4 - 0.12t^2$ Answer: -ve when $t = 10 \rightarrow$ slowing down Alternative for the above 3 marks: $v(10) = 0 \rightarrow$ slowing down	M1 A1 B1ft M1 A1 M1 A1 A1	For using $s(t) = \int v dt$ and attempting to integrate f.t. for wrong $s(t)$ For identifying the equation as a quadratic in t^2 and attempting to solve 5 For using $a = dv/dt$ and attempting to differentiate 3

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5	(i) $T_{AR} = T_{BR}$	B1	1	
	(ii) $T \cos RAB = T \cos RBA \rightarrow \text{angle } RAB = \text{angle } RBA$	B1	1	
	(iii) $2T \cos 60^\circ = 0.5g$	M1		For resolving forces on R vertically
	$T = 0.5g$	A1		May be implied
	$R = 0.3g + T \sin 30^\circ$	M1		For resolving the forces on B vertically (3 terms required)
	Answer: 5.5N	A1ft		f.t. for 3 + $\frac{1}{2}$ T
	Alternative for the above 4 marks:			
	For using $R_B = R_A + 0.3g$	B1		
	For resolving forces vertically on the whole system			
	($R_B + R_A = (0.5 + 0.3)g$) or for $R_A = \frac{1}{2}(0.5g)$ and eliminating R_A	M1		
	Answer: 5.5 N	A1		
	$T = 0.5g$	B1		
	$F = T \cos 30^\circ$	M1		For resolving the forces on B horizontally
	Answer: 4.33N	A1	6	
6	(i)	M1		For applying N2 to one particle, or for using $(m_1 + m_2)a = (m_1 - m_2)g$
	$0.5a = 0.5g - T$ or $0.4a = T - 0.4g$	A1		
	or $0.9a = 0.1g$			
		M1		For applying N2 to the other particle (if necessary) and solving for a
	Answer: 1.11ms^{-2}	A1	4	
	(ii) $v^2 = 2(g/9)4.5$	M1		For using $v^2 = 2as$
	$0 = g^{1/2} - gt$	M1		For using $0 = u + at$
	Answer: 0.316s	A1	3	
	(iii)	M1		For using distance is $2s$ and obtaining s from $(u + 0)/2 = s/t$, $0 = u^2 + 2as$ or $s = ut + \frac{1}{2}at^2$
	Answer: 1m	A1	2	

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7	(i)	1200g(500 sin6°) seen or implied	B1	Can be scored in 1 st or 2 nd part	
			M1	For using WD by driving force = PE gain + WD against resistance, or for using WD against resistance = (1800 – component of weight) × 500	
		1800 × 500 = 1200g(500 sin6°) + WD against resistance, or	A1ft	f.t. for wrong PE gain or equivalent	
		WD against resistance = (1800 – 1200gsin6°) × 500			
		Answer: 273 000J	A1	4	
	(ii)		M1	For using KE gain = $\frac{1}{2} m(v^2 - u^2)$	
		$\frac{1}{2} 1200(20^2 - 8^2)$	A1		
		WD = 201 600 + 627 170 + 700 × 500	M1	For using WD by driving force = KE gain + PE gain + WD against resistance	
		Answer: 1 180 000 J	A1	4	
	SR (For candidates who assume, implicitly or otherwise, that the acceleration is constant)				
(max 2 out of 4)					
For finding the acceleration (0.336) using $v^2 = u^2 + 2as$, applying Newton's 2 nd law to find the force of D's engine (2360) and multiplying by 500 to find the WD.					
		M1			
	Answer: 1 180 000 J	A1			
	(iii)		M1	For using $\frac{P_{top}}{P_{bottom}} = \frac{F_{top}}{F_{bottom}} \times \frac{v_{top}}{v_{bottom}}$	
		Ratio = 4 × 20/8	A1		
		Answer: 10	A1	3	
	SR (max 1 out of 3)				
For using calculated values of F in the ratio 4:1 (e.g. 2360 × 4 and 2360), and obtaining the answer 10:1 for required ratio.					
B1					